

Module 10: Epigenetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain why not all genes are turned on in every cell
2. Describe how the lac operon works
3. Define epigenetics and explain how it differs from mutations
4. Explain DNA methylation and histone acetylation
5. Give examples of how the environment affects gene expression

The Big Picture

Every cell in your body has the **same DNA**. A skin cell and a brain cell are genetically identical, yet they look and function completely differently. **Gene regulation** controls which genes are "on" in each cell type. **Epigenetics** is how these on/off patterns are set without changing the DNA itself.

Key Terms

Gene Regulation

- **Gene Expression** — Using a gene to make its protein product
- **Transcription Factor** — A protein that helps turn a gene on or off
- **Activator** — Turns a gene ON
- **Repressor** — Turns a gene OFF
- **Promoter** — The DNA spot where RNA polymerase binds to start reading
- **Enhancer** — A distant DNA spot that boosts transcription when activated
- **Housekeeping Gene** — A gene that's always on in every cell (e.g., genes for ribosomes)

The Lac Operon (Bacteria)

- **Operon** — A group of genes controlled by one switch (found in bacteria)
- **Lactose absent** → Repressor sits on operator → genes OFF (no enzymes made)
- **Lactose present** → Lactose changes repressor shape → repressor falls off → genes ON → enzymes made
- This is **inducible** — the default is OFF, and lactose turns it ON

Epigenetic Mechanisms

- **Epigenetics** — Changes in gene expression that do NOT change the DNA sequence
- **DNA Methylation** — Adding -CH₃ groups to DNA → gene turned **OFF** (silence)
- **Histone Acetylation** — Adding acetyl groups to histones → DNA loosens → gene turned **ON**
- **Euchromatin** — Loose, open DNA → genes CAN be read
- **Heterochromatin** — Tight, closed DNA → genes CANNOT be read
- **X-inactivation** — One X chromosome in females is shut down → Barr body
- **Calico cats** — Patches of color show random X-inactivation in different skin cells

Other Regulation

- **Alternative Splicing** — One gene can make different proteins by including different exons
- **miRNA** — A tiny RNA that blocks translation of specific mRNAs

Levels of Gene Control

Level	How It Works
Chromatin	Methylation/acetylation opens or closes DNA
Transcription	Transcription factors turn genes on/off
After transcription	Splicing, mRNA stability, miRNA
Translation	Blocking the ribosome
After translation	Modifying or destroying the protein

Environmental Effects on Genes

- **Diet** — Folic acid affects methylation patterns
- **Stress** — Can change which genes are active
- **Toxins** — Can disrupt normal epigenetic programming
- **Your genes are not your destiny** — Lifestyle can influence which genes are on or off

Study Tips

1. **Same DNA, different cells** — This is THE core concept of this module
2. **Acetylation = Activation** — Both start with "A"
3. **Lac operon logic** — No food? Don't waste energy making enzymes. Food arrives? Turn on the enzymes.
4. **Think about calico cats** — The best real-world example of epigenetics
5. **Epigenetics ≠ mutation** — Epigenetics changes the VOLUME, not the SONG